

# Evaluating Sparse, Dense, and Hybrid Retrieval for EU Climate Legal Documents

Rameez Wani  
School of Computing  
Dublin City University  
Dublin, Ireland  
rameezshafat.wani2@mail.dcu.ie

Ananya Warior  
School of Computing  
Dublin City University  
Dublin, Ireland  
ananyasalil.warior2@mail.dcu.ie

**Abstract**—Legal information retrieval in statutory corpora is hard because of formulaic language, hierarchical authority, and the need for doctrinal precision, and evaluating it fairly is harder still: existing hybrid legal-retrieval studies target case law or large general corpora, and none controls for query–corpus vocabulary leakage, where queries written while consulting the statutes inherit their wording so that term overlap stands in for relevance. This paper’s primary contribution is a leakage-controlled evaluation methodology for statute-level legal retrieval, together with evidence that evaluation design, more than retrieval architecture, shifts which approach appears superior on this corpus. The protocol has two parts: every query is rewritten in the plain language of a non-expert, and the test set is extended with information needs sourced from outside the corpus entirely, derived from written questions that Members of the European Parliament put to the Commission (31 of the 53 sealed test queries), whose vocabulary is independent of the statutes by construction. A paired experiment on 71 identical information needs isolates the effect: with corpus, indices, systems, and relevance judgements held fixed, rewording alone moves BM25’s MRR@5 by 0.29 against dense’s 0.09 (Section V-C), so leaky benchmarks manufacture a false parity between lexical and semantic retrieval. The methodology enables a controlled comparison of four first-stage configurations on a 102-query benchmark over EU Climate Law: BM25 lexical search, a learned-sparse encoder (SPLADE), `nomic-embed-text-v1.5` dense retrieval (768-dim, 8,192-token context), and a hybrid combining BM25 and `nomic-embed-text-v1.5` via weighted Reciprocal Rank Fusion (RRF,  $k=20$ , dense weight 5). The corpus comprises 1,156 article-level provisions (1,166 retrieval units after splitting oversized amendment articles) extracted from 72 EU legislative instruments via SPARQL and Formex XML; fusion parameters were tuned on a 49-query validation split and the sealed 53-query test set was evaluated only once. Under the controlled protocol, dense retrieval achieves  $\text{HR@5}=0.925$ , substantially outperforming BM25 ( $\text{HR@5}=0.698$ ; Wilcoxon  $p=0.004$ ). SPLADE closes most of the gap to dense ( $\text{HR@5}=0.887$ ,  $\text{MRR@5}=0.744$ ,  $\text{NDCG@5}=0.717$ ) while carrying the highest hard-negative rate of any system, so classical BM25 understates what sparse retrieval can do here. The hybrid achieves  $\text{HR@5}=0.906$ ,  $\text{MRR@5}=0.773$ , and  $\text{NDCG@5}=0.721$ , the highest MRR and NDCG of the three primary systems, though SPLADE’s point estimates are close behind and the hybrid’s improvement over dense alone is not significant ( $p=0.614$ ); only 5 of 53 queries (9.4%) are hard failures for the hybrid. An embedding-model ablation adds a null result: E5-large-v2 and BGE-small-v1.5, both restricted to 512-token windows that truncate a fifth of this corpus, show no significant difference from the 8,192-token encoder (Wilcoxon  $p\geq 0.055$ ), and a targeted check finds the document-chunking strategy makes no measurable

difference either (Section V-G). The transferable finding is methodological: on a leakage-controlled benchmark a strong general-domain dense retriever already performs close to the best point estimate any system reaches on this corpus, and weighted lexical-semantic fusion adds modest ranking gains without a statistically significant coverage improvement, so we argue leakage control and externally sourced information needs should be standard practice when benchmarking statute-level retrieval.

**Index Terms**—Legal Information Retrieval, Hybrid Retrieval, BM25, Dense Retrieval, Reciprocal Rank Fusion, EU Climate Law, Retrieval-Augmented Generation, Statute-Level Benchmark

## I. INTRODUCTION

Primary legislation presents significant challenges for legal interpretation, particularly in identifying the relevant document, which depends on precise wording, intertextual references, and correctly identifying the applicable instrument among multiple similarly titled sources. These challenges are especially pronounced in EU climate and energy law: the Emissions Trading System directive of 2003 has been amended repeatedly, the Carbon Border Adjustment Mechanism borrows its allowance definitions from that directive, and dozens of implementing and delegated acts reuse the same vocabulary (*free allocation, carbon leakage, monitoring and reporting*) for legally distinct obligations. A retrieval system that returns a plausible but wrong instrument does not produce an approximate answer; it produces the wrong law. (Our evaluation, described in Section IV, judges relevance at this instrument level rather than the finer article level; Section VII discusses what that leaves open.) The stakes are highest in retrieval-augmented generation (RAG) pipelines, where the first-stage retriever determines whether citation-grounded generation is possible at all [13]. This paper evaluates that first stage.

The standard tension in legal retrieval is between lexical and semantic matching, but which side of it wins is a claim about a benchmark as much as about a system. BM25 performs well on formulaic legal text, where exact term overlap is a strong relevance signal [1], but a compliance officer asking whether her shipping company “has to pay for the pollution its vessels cause” shares almost no vocabulary with “surrender requirements in respect of maritime transport activities” (Directive 2023/959). Dense retrievers [19] bridge that vocabulary

gap, but bring failure modes of their own: semantically similar passages from the wrong instrument can rank highly because legal drafting recycles structural boilerplate [11], and off-the-shelf embedding models often degrade on specialised corpora outside their training distribution [13], [18]. Hybrid sparse–dense fusion is the commonly recommended remedy [2], [3], [9]. Whether that recommendation holds for statute-level retrieval over a small, terminologically dense corpus is an empirical question, and answering it fairly depends on a benchmark whose queries were not written while consulting the statutes: when they are, they inherit statutory wording, term overlap substitutes for relevance, and lexical and hybrid systems are flattered by construction. No prior hybrid legal-retrieval study controls for this.

Our first evaluation attempt showed this concretely. In a pilot benchmark whose queries were written while consulting the corpus, every configuration looked excellent, the hybrid appeared to win outright, and the strongest system reached a perfect hit rate; the queries had simply inherited the statutes’ vocabulary, so the benchmark rewarded term overlap rather than retrieval quality and flattered BM25 in particular. We therefore rebuilt the benchmark under an explicit leakage-control protocol: every query was rewritten in the plain language of a non-expert, and the test set was extended with queries derived from written questions that Members of the European Parliament submitted to the Commission, real information needs documented by their reference numbers, whose official answers often name the very instruments a correct retrieval should return. Because those needs originate outside the corpus, their vocabulary is independent of the statutes by construction rather than by rewriting. To our knowledge no prior work on hybrid legal retrieval has evaluated statute-level EU climate law, and none has controlled for query–corpus vocabulary leakage this way; we argue the protocol, plain-language rewriting plus externally sourced information needs, should be standard practice for statute-level retrieval evaluation.

We ask three questions. **RQ1.** Does hybrid sparse–dense retrieval improve effectiveness and reliability over BM25-only and dense-only baselines for EU Climate Law statute retrieval? **RQ2.** What failure modes remain for the strongest configuration, and what do they suggest about where retrieval effort should go next? (This paper evaluates retrieval only; it does not measure generation, so it cannot establish whether retrieval is the bottleneck for a full RAG pipeline.) **RQ3.** Does dense embedding-model choice on this corpus track context-window length, or must it be benchmarked directly?

Our contributions are: (i) a leakage-control protocol for statute-level retrieval evaluation, pairing plain-language rewriting of every query with a test set extended by externally sourced information needs, plus a paired experiment quantifying what uncontrolled leakage costs (Section V-C); (ii) an externally sourced query set: 31 of the 53 sealed test queries derived from documented European Parliament written questions, with vocabulary independent of the corpus by construction; (iii) a 102-query benchmark for statute-level EU

Climate Law retrieval with binary CELEX-level relevance judgements built under that protocol, over a reproducible corpus of 1,156 article-level provisions from 72 EU climate instruments extracted through a documented two-phase SPARQL pipeline over CELLAR; (iv) a comparison of BM25, a learned-sparse encoder (SPLADE), a pure dense bi-encoder, and weighted reciprocal rank fusion, with paired significance tests (Wilcoxon, exact McNemar), bootstrap and cluster-bootstrap confidence intervals, and per-query failure analysis; and (v) an embedding-model ablation (E5-large-v2, BGE-small-v1.5) showing that context-window length does not measurably predict retrieval quality on this corpus, together with a chunking-strategy check and a BM25 hyperparameter search run under the same validation/test discipline as the RRF tuning.

To preview the results: on the leakage-controlled test set both the dense retriever and the hybrid significantly outperform BM25 but are statistically indistinguishable from each other, and the direction of that comparison reversed as the test set grew, a caution against reading small-sample differences between strong systems. The residual failures are ranking errors rather than coverage gaps, the largest a cross-instrument discrimination error that a different open-weight encoder resolves from the same indexed text, pointing to embedding-model choice rather than corpus preprocessing as the next lever for improvement.

Section II reviews related work; Section III details the corpus, retrievers, and fusion; Section IV covers the benchmark and setup; Section V reports results; Section VI discusses implications; Section VII concludes with limitations and future work.

## II. RELATED WORK

### A. Lexical vs. Dense Retrieval in Legal IR

Mori et al. [1] find BM25 a competitive baseline for formulaic legal text, where dense models underperform by glossing over legally binding phrasing; Zheng et al. [7] find the opposite on reasoning-intensive queries, and LRAGE [13] confirms off-the-shelf dense retrievers underperform BM25 without domain adaptation. Li et al. [21] add that on large-scale Chinese legal case retrieval BM25 consistently beats off-the-shelf dense models, so the balance is corpus- and language-dependent; our results suggest it also depends on how queries are phrased.

### B. Hybrid Retrieval

Rayo et al. [3], LegalRAG [9], and HyPA-RAG [2] each show hybrid fusion beating single-mode retrieval on regulatory or multilingual legal text. SPLADE [4] learns sparse representations that close the vocabulary gap inside a single inverted index; we deploy classical BM25 as the fusion partner for interpretability but add SPLADE as a standalone baseline (Section V-A) to check how much headroom that leaves on the table. Louis et al. [25] find domain-general legal models help fusion zero-shot but can hurt once components are domain-trained unless weights are re-tuned, consistent with our finding

that an untuned literature-default RRF configuration underperforms a corpus-specific one; no legal-domain or EU-law-specific embedding model is evaluated here (Section VII). Cormack et al. [14] established that rank-based fusion beats score-based combination without score normalisation; Bruch et al. [20] show analytically that RRF’s parameters do not transfer across collections, which our validation sweep reproduces on a much smaller corpus.

### C. Retrieval Reliability and Legal RAG

Reuter et al. [11] name Document-Level Retrieval Mismatch (DRM), where shared structural boilerplate makes globally irrelevant documents rank highly. Ho et al. [12] argue doctrinal accuracy needs structural signals beyond semantic similarity; CBR-RAG [5] and LexRAG [6] show multi-turn legal grounding needs retrieval robust to topic drift. Multi-stage embedding-plus-LLM-reranking works well in COLIEE [10], and LexDrafter [8] applies RAG directly to EU legislative drafting, the closest prior work to our domain, though it does not evaluate first-stage retrieval quality.

## III. METHODOLOGY

The pipeline is a two-phase CELLAR extraction feeding dual BM25/dense indexing, with both retrievers’ outputs merged at query time by weighted RRF; each stage is detailed in the subsections below.

### A. Corpus Construction

We extract the corpus from the EU CELLAR repository in two phases. A SPARQL query first retrieves every work URI associated with any of 21 EuroVoc climate and energy concept identifiers, which yields the 72 instruments; per-document manifestation queries then resolve the Formex XML (`fmxml`) download URL for each instrument, falling back to the HTML manifestation for the few documents that lack one. Formex is the Publications Office’s structured drafting format, so parsing it (with `lxml`) recovers the legal structure of each act directly instead of reverse-engineering it from rendered HTML.

The retrieval unit is the legislative article, a deliberate choice: EU articles are typically drafted to state one obligation, definition, or procedure, so article-level units align with what a legal user needs retrieved and avoid the arbitrary window sizes of token-based chunking. We have not measured how often an answer instead spans multiple articles, annexes, recitals, or cross-referenced instruments, so this is motivated by drafting convention rather than empirically verified, and Section V-G’s chunking check speaks only to the five oversized amendment articles, not this broader question. Concretely, each `<ARTICLE>` element becomes one unit, with text reconstructed from `<ALINEA>` leaf nodes to avoid duplicated nesting; every record is validated against a schema, and placeholder articles are discarded. The exception is amending instruments, whose Article 1 can run to tens of thousands of tokens because it contains every amendment the act makes. Five such articles exceed a 7,000-token limit (chosen to leave headroom below the dense encoder’s 8,192-token window)

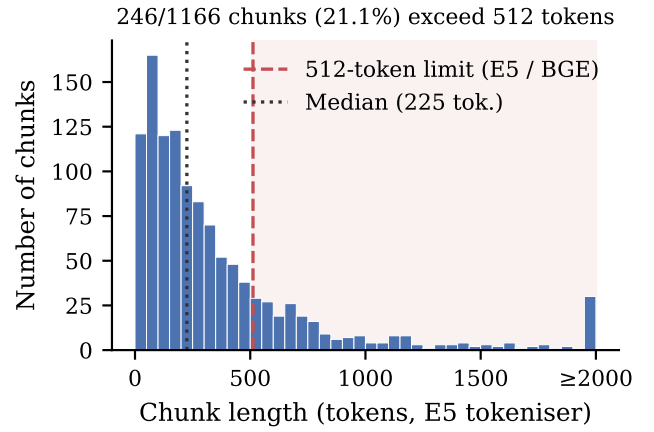


Fig. 1: Token-length distribution of the 1,166 retrieval units (E5 tokeniser; median 225 tokens). 246 units (21.1%) exceed the 512-token window of a conventional encoder (dashed line), the text a short-context model silently truncates; the deployed 8,192-token encoder reads every unit whole.

and are split at the natural amendment-point boundaries; the resulting sub-chunks inherit the parent `celex_id`, so retrieval results always resolve to a citable instrument. Fig. 1 shows the resulting length distribution.

The final corpus contains **1,156** article-level provisions (**1,166** retrieval units after splitting) from **72** CELEX instruments covering ETS, CBAM, Taxonomy, LULUCF, F-Gas, Maritime MRV, ESR, the European Climate Law, Energy Efficiency, CSDDD, and the Governance Regulation. Instrument size is skewed, as is typical of EU legislation: a median of 10 articles per instrument against a mean of 16.2, ranging from 2 (a short amending decision) to 72 (the European Green Bond Standards Regulation), and 79.8 % of retrieval units come from Regulations against 20.2 % from Directives. The 21 EuroVoc identifiers and all 72 CELEX instrument IDs are released with the code rather than listed here; we have not obtained an independent legal expert’s completeness assessment, so coverage rests on the EuroVoc concepts alone. Indexing extracts only `<ARTICLE>` elements; annexes, recitals, tables, and footnotes are not indexed or retrievable.

### B. Sparse Retrieval - BM25

The deployed lexical retriever is BM25Okapi [15] (`rank-bm25` 0.2.2) with  $k_1=1.5$  and  $b=0.75$ , checked rather than assumed: the same validation grid search used for RRF tuning prefers  $k_1=1.8$ – $2.0$ ,  $b=1.0$ , but that configuration’s hybrid has *lower* test-set MRR@5 (0.737 vs. 0.773) and a worse hard-negative rate (0.30 vs. 0.20), so we keep the defaults rather than switch on an overfit signal. Tokenisation preserves hyphenated terms of art (*net-zero*) as single tokens and applies neither stemming nor stopword removal, for precision on differently scoped EU terms and fixed phrases (*free of charge*); this costs real performance (+0.057 HR@5, +0.058 MRR@5 available from stemming alone), a trade we disclose

rather than hide. At query time BM25 scores all 1,166 units and returns the top 100.

### C. Learned-Sparse Retrieval - SPLADE

To test whether classical BM25 understates what sparse retrieval can do here, we add `naver/splade-cocondenser-ensembledistil` as a standalone baseline (not fused into the hybrid). SPLADE encodes both queries and documents into sparse vectors over the tokenizer vocabulary via a masked-language-model head, term weights are  $\log(1+\text{relu}(x))$ -pooled over token positions, giving implicit query expansion while remaining sparse and interpretable. At this corpus size (1,166 documents) we score by exact sparse dot product rather than an inverted-index engine such as Anserini/Lucene. The trade is deliberate: brute-force scoring carries no approximate-retrieval error, so the SPLADE numbers below reflect the model and not an index-pruning heuristic, but they equally say nothing about learned-sparse behaviour, or its engineering overhead, at the corpus scales where an inverted index becomes necessary. Indexed on the same hierarchically chunked corpus as the other retrievers, for comparability.

### D. Dense Retrieval - nomic-embed-text-v1.5

The semantic retriever is `nomic-embed-text-v1.5` [16], a 768-dimensional bi-encoder with an 8,192-token context window, loaded via `sentence-transformers`. Three properties drove the selection. It is fully open-weight, which keeps the pipeline reproducible without API access. Its long context window embeds EU articles whole (only the five oversized amendment articles described above need splitting), where a conventional 512-token encoder would silently truncate roughly a fifth of this corpus (Fig. 1); Section V-G tests how much this actually matters. Most importantly for the experimental design, it is a *pure* bi-encoder: models trained jointly for dense, sparse, and late-interaction retrieval (such as BGE-M3) already encode lexical signals in their dense vectors, which would muddy any conclusion about what BM25 adds on top.

Following the model’s asymmetric training scheme, queries are prefixed with `"search_query:"` and articles with `"search_document:"` before encoding. Corpus vectors are L2-normalised and stored in a FAISS [17] `IndexFlatIP`, where inner product on normalised vectors equals cosine similarity; at 1,166 vectors, exact search is trivially fast and no approximate index is needed. At query time, FAISS returns the top 100 nearest neighbours.

### E. Reciprocal Rank Fusion

Both retrievers run concurrently on each query (`ThreadPoolExecutor`, two workers), and their ranked lists are merged by weighted Reciprocal Rank Fusion [14]. Each candidate document  $d$  receives the score

$$\text{RRF}(d) = \frac{w_s}{k + \text{rank}_s(d)} + \frac{w_d}{k + \text{rank}_d(d)} \quad (1)$$

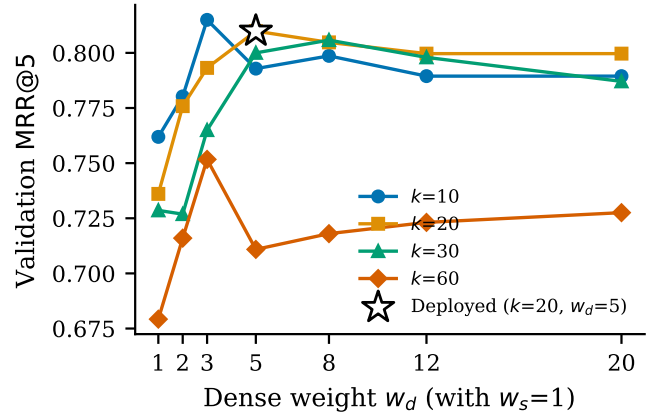


Fig. 2: Validation-split MRR@5 across the full  $(k, w_d)$  grid at the deployed retrieval budget (top- $k_r=100$ ),  $k \in \{10, 20, 30, 60\}$  matching the documented sweep. The deployed configuration ( $k=20$ ,  $w_d=5$ ; MRR@5=0.810) is the best-ranking cell that also reaches HR@5=1.000; the literature-default region ( $k=60$ , low  $w_d$ ) performs worst throughout.

where  $\text{rank}_s$  and  $\text{rank}_d$  are the document’s positions in the sparse and dense lists ( $\infty$  if absent), and  $w_s$ ,  $w_d$  weight the two retrievers. The constant  $k$  is a positive smoothing term added to each rank in the denominator; it sets how steeply a document’s contribution decays with rank position, so a small  $k$  lets rank-1 results dominate the fused score while a large  $k$  flattens the differences between neighbouring ranks. The intuition of (1) is that a document ranked well by both retrievers should beat a document ranked first by only one; the weights let a systematically stronger retriever dominate ties. The top 5 fused results form the system’s output (and, in a full RAG deployment, the context passed to generation).

We fuse ranks rather than raw scores because the two score scales are incompatible: BM25 produces unbounded positive reals while cosine similarity lies in  $[-1, 1]$ , and any direct combination smuggles in scale assumptions that rank-based fusion avoids entirely [14]. The operating point  $k=20$ ,  $w_s=1$ ,  $w_d=5$  was chosen by grid search on the 49-query validation split; the sealed 53-query test set was evaluated only after these values were frozen. Fig. 2 shows the validation grid: ranking quality improves as the dense retriever is up-weighted, degrades sharply at large  $k$ , and the deployed configuration is the highest-MRR cell among those reaching HR@5=1.000 on validation; this “highest-MRR-among-perfect-coverage” selection rule was applied after inspecting the grid rather than pre-registered beforehand, a source of model-selection bias we disclose rather than correct for post hoc, since the grid itself (Fig. 2) shows a broad region of comparable cells rather than one selection-sensitive spike.

## IV. EXPERIMENTAL SETUP

### A. Benchmark

The benchmark contains 102 queries covering all eleven EU Climate Law instrument families, constructed in two

stages. Unlike broad multi-task legal benchmarks that aggregate hundreds of tasks across jurisdictions [22], it targets a single tightly bounded corpus, which is what makes statute-level retrieval evaluation and leakage control tractable. An initial set of 71 queries was written in deliberately natural, non-statutory language: an early pilot in which queries paraphrased corpus wording produced near-perfect scores for all systems (HR@5=1.0), indicating query–corpus vocabulary leakage rather than retrieval quality; all queries were therefore rewritten to express the underlying information need without reusing statutory vocabulary. These 71 queries were split 70/30 by stratified difficulty (seed 42) into a 49-query validation set (used exclusively for fusion-parameter tuning) and a 22-query test set. The paired leakage experiment of Section V-C operates on exactly this original 71-query set (49 validation plus 22 test), because only these queries exist in both wordings: the corpus-derived phrasing of the pilot and the plain-language rewrite. The test set was subsequently expanded with 31 queries derived from written questions submitted by Members of the European Parliament to the Commission: each parliamentary question was screened for answerability from legislation, paraphrased into plain language, and mapped to its relevant CELEX instruments, with the parliamentary reference number (e.g., E-002271/2024) recorded per query as an audit trail. In many cases the Commission’s official answer cites the relevant instruments, providing independent corroboration of the relevance judgement. Corpus membership was verified for every mapping before a query was accepted, so no query requires an instrument absent from the corpus. These 31 queries take no part in the paired leakage experiment, since they never had a corpus-derived wording to be paired with; that is also their value. Because they originate outside the corpus, their vocabulary is independent of the statutes by construction rather than by rewriting, which makes them the strongest leakage control in the benchmark, and each is traceable to a public parliamentary record. Rewriting still lost something: re-running retrieval on 29 of these 31 queries’ literal original text (2 reference pre-2024 terms not retrievable via the current register) gives BM25 +0.207 HR@5 and +0.271 MRR@5 over the paraphrase, with dense essentially flat, corroborating the paired leakage experiment of Section V-C on data it never touched. Qrels were built by a single author, not pooled across systems; a post-hoc pooling diagnostic (union of all five systems’ top-10 hits) finds candidates outside the recorded qrels for every query (mean 10.7), a possible annotation gap we quantify but a human would need to resolve. All reported results use the resulting sealed **53-query test set**; the validation set was never evaluated against reported metrics. To give a flavour of the derivation: parliamentary question E-004153/2025 asks whether a ship calling at a mainland port en route to an outermost region owes allowances for the first leg of the voyage; this becomes the plain-language query “A ship sails from the mainland, stops at another mainland port, then continues to one of the EU’s remote overseas regions — is the whole journey spared the carbon charge, or only the last leg?”, mapped to Directive 2023/959, whose derogation

TABLE I: Benchmark composition. “ $\geq 2$  rel.” counts queries with two or more relevant instruments; “instr.” counts distinct instruments used as relevance targets.

| Subset                | $n$ | standard | hard | $\geq 2$ rel. | instr. |
|-----------------------|-----|----------|------|---------------|--------|
| Validation            | 49  | 36       | 13   | 29            | 44     |
| Test (original split) | 22  | 16       | 6    | 9             | 26     |
| Test (EP-derived)     | 31  | 19       | 12   | 13            | 22     |
| Test (total)          | 53  | 35       | 18   | 22            | 37     |
| All queries           | 102 | 71       | 31   | 51            | 62     |

the Commission’s answer cites.

Each query carries a difficulty tag used for stratification: *hard* queries share essentially no vocabulary with the statutory text (e.g., “the don’t-make-anything-else-worse test” for the Taxonomy’s *do no significant harm* principle), while *standard* queries are naturally phrased but moderately terminological. Table I summarises the composition. Relevance judgements target 62 of the 72 corpus instruments; the ten never used as targets are annex-replacement amendment acts with no self-contained content, excluded as relevance targets by construction.

Relevance is binary at the CELEX instrument level: a document is relevant if it contains the provision that directly and materially answers the query. CELEX identifiers with corrigendum suffixes (e.g., 32003L0087R(02)) are normalised by stripping the R(. . .) suffix before matching.

## B. Metrics

We report three metrics at cut-off 5, matching the number of provisions a downstream RAG stage would receive as context. HR@5 measures coverage: the fraction of queries for which at least one relevant instrument appears in the top 5 at all. For grounded generation this is the make-or-break number: a miss here cannot be repaired downstream. MRR@5 measures how high the first relevant result lands, which matters because generation quality degrades when the correct provision sits below distractors. NDCG@5 credits finding *all* relevant instruments, not just the first: 22 of the 53 test queries have two or more relevant instruments (Table I), and MRR is blind to everything after the first hit. Gains are binary; since retrieval is article-level, several articles of one instrument can occupy the top 5, so truncation happens first and deduplication second, crediting each relevant instrument once at its highest-ranked occurrence (duplicate slots are not backfilled).

With only 53 test queries we quantify uncertainty several ways: 95% CIs for all three metrics by percentile bootstrap (5,000 resamples, seed 42) and by a cluster bootstrap resampling target instruments rather than queries (comparably wide on this benchmark); exact McNemar tests for the binary HR@5 comparisons and Wilcoxon signed-rank for MRR@5/NDCG@5 ( $\alpha=0.05$ ,  $n=53$ ), with Cohen’s  $d$ , a matched-pairs rank-biserial correlation, and Holm-Bonferroni

correction across the six embedding-ablation tests (none significant before or after).

### C. Hardware and Software

All results were produced on consumer hardware, no GPUs or paid APIs: primary systems on an AMD Ryzen 7 5700U laptop (CPU-only, Windows 11) and the embedding ablation on an Apple M1 Pro (CPU-only), with primary numbers reproduced identically on both. The software stack is Python 3.14.2 with sentence-transformers 5.5.1, faiss-cpu 1.14.2, rank-bm25 0.2.2, and pydantic 2.13.4, all pinned in the repository; the combined indices occupy under 10 MB. All four dense/sparse encoders (nomic-v1.5, E5-large-v2, BGE-small-v1.5, SPLADE) share batch size 64 (16 for SPLADE) at PyTorch’s CPU default precision (fp32); query/document prefixes are given with each model in Section III.

## V. PERFORMANCE EVALUATION

### A. Overall Effectiveness

Fig. 3 reports the headline comparison on the sealed 53-query test set: hybrid vs. BM25  $p=0.0005$ ; dense vs. BM25  $p=0.004$ .

Both semantic configurations beat BM25 by more than twenty points of HR@5 (hybrid:  $p=0.0005$ ,  $d=0.579$ ; dense:  $p=0.004$ ,  $d=0.433$ ): on a leakage-controlled benchmark, lexical retrieval alone still misses roughly three queries in ten. SPLADE, a standalone baseline rather than BM25’s replacement, closes most of that gap: its point estimates sit close to nomic-embed-text-v1.5’s and well above BM25’s, above the hybrid’s on no metric, while carrying the highest hard-negative rate of any system tested (0.30).

Between dense and the hybrid the Wilcoxon test finds no significant difference ( $p=0.614$ ,  $d=0.120$ ): dense posts the best HR@5, the hybrid the best MRR@5 and NDCG@5, and their confidence intervals overlap substantially on every metric (Fig. 3). Neither reading is confirmed equivalence, only that this test, at this sample size, detected no difference. Part of the explanation is architectural: RRF fuses ranks and discards the scores behind them, so it cannot weight the dense retriever’s confidence, and with dense already at rank 1 on 34 of 53 queries (Section V-F) BM25’s ranks are as likely to displace a correct dense result as to confirm one (Section V-E), consistent with Bruch et al. [20]. The honest answer to RQ1 is therefore split: fusion delivers a significant improvement over the lexical baseline but no measurable improvement over a strong dense retriever alone.

### B. Effect of Query Difficulty

Table II splits the test set by the difficulty tag introduced in Section IV: *standard* queries are naturally phrased but moderately terminological, while *hard* queries share essentially no vocabulary with the statutes.

The split localises where lexical retrieval breaks: BM25’s MRR@5 nearly halves on hard queries (0.611 to 0.366) while dense degrades far more gently (0.784 to 0.661), the gap

TABLE II: Results by query difficulty (standard  $n=35$ , hard  $n=18$ ). nomic=nomic-embed-text-v1.5; Hybrid=weighted RRF; BGE=BGE-small-v1.5. SPLADE and BGE are standalone baselines, not fused into the hybrid.

| System | Subset   | HR@5  | MRR@5 | NDCG@5 |
|--------|----------|-------|-------|--------|
| BM25   | standard | 0.743 | 0.611 | 0.559  |
|        | hard     | 0.611 | 0.366 | 0.367  |
| nomic  | standard | 0.943 | 0.784 | 0.738  |
|        | hard     | 0.889 | 0.661 | 0.654  |
| Hybrid | standard | 0.943 | 0.821 | 0.754  |
|        | hard     | 0.833 | 0.680 | 0.655  |
| SPLADE | standard | 0.943 | 0.796 | 0.778  |
|        | hard     | 0.778 | 0.644 | 0.598  |
| BGE    | standard | 0.943 | 0.746 | 0.704  |
|        | hard     | 0.944 | 0.704 | 0.680  |

widening from 0.17 to 0.30, direct evidence that BM25’s failure mode is vocabulary, not ranking. It also explains where fusion helps at all: the hybrid’s MRR@5 edge over dense is concentrated on standard queries (+0.037) and shrinks on hard ones (+0.019), since BM25’s signal is only trustworthy when the query shares some statutory vocabulary.

The learned-sparse and small-dense baselines split along the same line. On standard queries all four semantic systems reach identical coverage (HR@5=0.943), and SPLADE posts the highest NDCG@5 (0.778): it is fully competitive when the query shares statutory vocabulary. On hard queries SPLADE drops further than dense, its HR@5 falling 0.165 from the standard split against dense’s 0.054, landing between BM25 and the semantic systems, so a learned-sparse model still carries a residual lexical dependence. BGE-small-v1.5 is nearly flat across difficulty and posts the best hard-query coverage of any system (HR@5=0.944), including the q080 instrument every primary system misses (Section V-G); neither effect shows in the aggregate ranking.

### C. Quantifying Vocabulary Leakage

The pilot benchmark’s original corpus-derived queries were preserved, which permits a controlled experiment: the same 71 information needs, with identical relevance judgements, evaluated under both their original statutory wording and their leakage-controlled rewriting. Nothing else varies: corpus, indices, and systems are held fixed. Table III reports the result.

Under leaky wording, all three systems sit within three points of one another and the benchmark cannot distinguish them. Rewriting the queries costs BM25 0.254 of HR@5 and 0.290 of MRR@5, while the dense retriever loses only 0.028 and 0.085 respectively; leakage inflates lexical retrieval roughly three and a half times more than semantic retrieval: corpus-derived wording disproportionately increases BM25’s apparent effectiveness in this benchmark, producing a near-parity that does not survive realistic phrasing. Two caveats: these 71 pairs include the 49 validation queries, so the absolute values are not comparable with Fig. 3 and carry no held-out



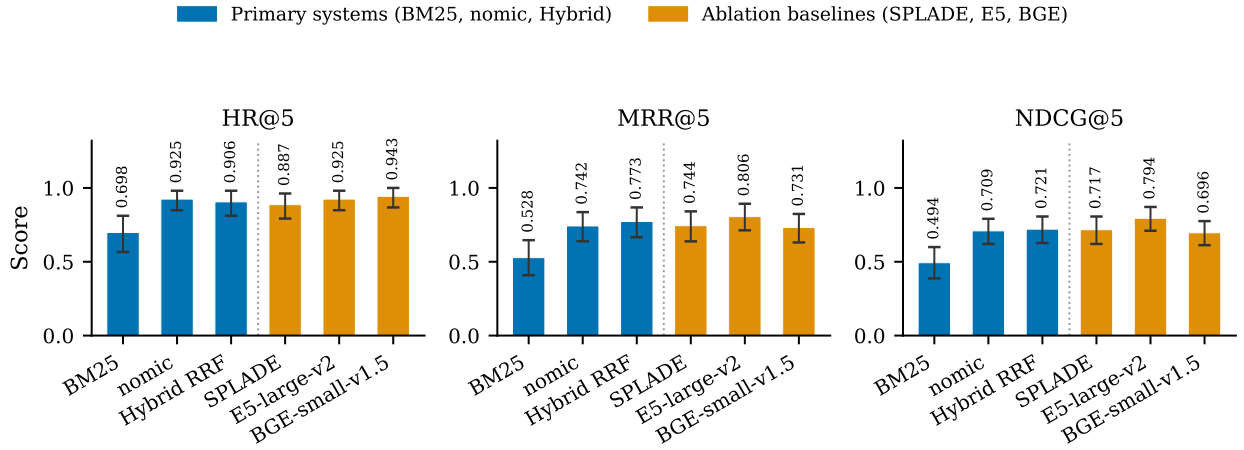


Fig. 3: HR@5, MRR@5, NDCG@5 for all six systems, with 95 % bootstrap CIs. Primary systems (blue) and ablation baselines (orange) sit on either side of the dotted divider.

TABLE III: Paired leakage experiment: 71 identical information needs and relevance judgements under corpus-derived (leaky) versus plain-language (controlled) wording.

| System                | Leaky wording |       | Controlled wording |       |
|-----------------------|---------------|-------|--------------------|-------|
|                       | HR@5          | MRR@5 | HR@5               | MRR@5 |
| BM25                  | 0.958         | 0.846 | 0.704              | 0.556 |
| nomic-embed-text-v1.5 | 0.986         | 0.870 | 0.958              | 0.785 |
| Hybrid RRF            | 0.986         | 0.876 | 0.972              | 0.795 |

guarantee; and no tuning was performed on either wording of this set beyond the validation use already described. The comparison is between wordings, not systems, and for that purpose the pairing is exact. SPLADE and BGE-small-v1.5 are absent here by construction: added as standalone baselines after the pilot, they never had a corpus-derived wording to pair against.

#### D. A Worked Example

Table IV shows the actual top-5 lists for one hard test query: “*Is there any leeway in the EU’s land-carbon rules for a country whose forests get wrecked by wildfires or bark beetles?*” The relevant instruments are the LULUCF Regulation (2018/841) and its 2023 amendment (2023/839), whose flexibility provisions speak of “natural disturbances”; none of the query’s content words appears in the statute. With nothing lexical to anchor on, BM25 is confidently wrong across two unrelated regimes, filling its list with the European Green Bond Standard and CBAM instruments on the strength of incidental term matches. The dense retriever maps the paraphrase onto the natural-disturbance provisions directly, placing both relevant instruments in its top five; the dense-weighted fusion preserves them. This single example illustrates the dominant pattern rather than proving a mechanism; Section V-E gives the full per-query failure breakdown, including the two fusion-weighting artefacts (q054, q090) and the one genuine coverage gap (q083) it does not show here. The deployed weighting

TABLE IV: Top-5 for the wildfire/bark-beetle query. Relevant in **bold**; repeats are different articles of one instrument.

| Rk | BM25        | nomic-embed-text-v1.5 | Hybrid           |
|----|-------------|-----------------------|------------------|
| 1  | EuGB’23     | <b>LULUCF’23</b>      | <b>LULUCF’23</b> |
| 2  | CBAM am.’25 | <b>LULUCF’23</b>      | <b>LULUCF’23</b> |
| 3  | CBAM’23     | <b>LULUCF’18</b>      | Nat.Rest.’24     |
| 4  | CBAM’23     | Nat.Rest.’24          | <b>LULUCF’18</b> |
| 5  | EuGB’23     | <b>LULUCF’18</b>      | <b>LULUCF’18</b> |

TABLE V: Per-query hit pattern breakdown ( $n=53$ ).

| Category                          | $n$ | Fraction |
|-----------------------------------|-----|----------|
| All three systems hit             | 36  | 0.679    |
| Dense and hybrid hit; BM25 misses | 12  | 0.226    |
| Only BM25 hits                    | 1   | 0.019    |
| Only dense hits (hybrid misses)   | 1   | 0.019    |
| All three miss                    | 3   | 0.057    |

is dense-dominated weighted RRF ( $w_d=5w_s$ ), not a balanced hybrid, and its behaviour should not be generalised to hybrid retrieval broadly (Section VI).

#### E. Hard Failure Analysis

RQ2 asks what failures remain; Table V breaks down per-query outcomes across the three systems.

Dense retrieval recovers 13 queries that BM25 misses entirely, confirming that semantic generalisation is the primary source of improvement over lexical retrieval. Fusion is nearly

lossless with respect to the dense ranking: the hybrid keeps 48 of dense’s 49 hits, losing one query to BM25 interference, and BM25 contributes one unique hit that the dense-favouring weight ( $w_d=5$ ) then suppresses. Three queries are missed by all three systems and the hybrid fails five in total; in every case the relevant instrument is present in the corpus, so these are ranking failures, not coverage gaps. The five split into four causes. q080 is a cross-instrument discrimination failure: the relevant ETS2 provisions are cleanly indexed (BM25’s best chunk for the gold instrument sits at rank 13) but both retrievers rank thematically adjacent instruments higher on shared vocabulary, and a different encoder retrieves the same text correctly (Section V-G), so the failure is a property of the deployed embedding model, not the corpus. q083’s content (Member State opt-in of fishing vessels and inland navigation) is confirmed absent from the codified text. q009’s colloquial phrasing sits too far from the statutory register of due-diligence obligations for either signal. q054 and q090 are fusion-weighting artefacts, one retriever placing the document correctly and the dense-heavy weighting overriding it (for q054, BM25 alone has the answer at rank 1). An earlier diagnosis blaming amendment-act sub-chunking was disproved by inspecting the indexed chunks, which matters because re-chunking would not have helped. Answering RQ2: the residual failures are concentrated in encoder-specific discrimination and extreme vocabulary mismatch, not in corpus coverage or the retrieval strategy.

#### F. Ranking Quality

Where does the hybrid’s aggregate MRR@5 advantage (+0.031) come from? Restricting to the 48 queries both semantic systems hit, as such analyses typically do, would exclude the 5 queries where dense and hybrid disagree on coverage, precisely those most informative about the HR@5 gap, so we report the full 53-query breakdown. On coverage, both hit on 48, dense alone on 1 (q090, pushed just outside the top 5 by fusion), hybrid alone on 0, and both miss 4. On ranking, over the 48 queries where the comparison is defined, the two place the first relevant result at the same rank on 31, the hybrid higher on 8, and dense higher on 9. The aggregate edge is one of magnitude: the hybrid more often promotes a relevant instrument all the way to rank 1 (38 queries versus 34 for dense and 23 for BM25), typically where BM25’s exact-term signal confirms an instrument `nomic-embed-text-v1.5` placed just below a semantically similar alternative. This is a post hoc exploratory reading, not a pre-specified mechanism; at this corpus size it moves the mean without reaching significance.

#### G. Embedding-Model Ablation

`nomic-embed-text-v1.5`’s context window was part of its selection (Section III), so we tested how much that choice matters, re-running dense-only evaluation with two alternative open-weight bi-encoders, E5-large-v2 [23] (335M params, 512-token window) and BGE-small-v1.5 [24] (33M params, same window), both indexed on the 1,156 raw articles and both silently truncating the 21.1 % of units exceeding 512

tokens (Fig. 1); scores in Fig. 3. Neither differs significantly from `nomic-embed-text-v1.5` (six Wilcoxon comparisons,  $p=0.055$ – $1.0$ ). E5-large-v2 posts the best MRR/NDCG point estimates of any system (MRR@5=0.806,  $p=0.216$ ; NDCG@5=0.794,  $p=0.055$ ) on a different query subset than `nomic-embed-text-v1.5` (49/53 agreement). BGE-small-v1.5, a fraction of the parameter count, posts the best coverage (HR@5=0.943,  $p=0.564$ ), retrieving q080, which all three primary systems miss, from exactly the same indexed text (Section V-E). Two readings, kept apart: context-window length does not predict retrieval quality here, models truncating a fifth of the corpus are statistically indistinguishable from one reading it whole, so the truncation argument for model selection carries less weight than assumed at design time; but these are null results at  $n=53$ , failing to detect a difference is not evidence of equivalence, and we claim parity only in that weak sense. A targeted check isolates chunking from encoder choice: re-indexing `nomic-embed-text-v1.5` with the five oversized Article 1 instruments left whole (truncated at encoding time, not split) reproduces HR@5, MRR@5, and NDCG@5 to four decimal places across all 15 affected test queries, including q080, direct evidence, not just chunk inspection, that chunking is not the cause of that failure. This is one alternative on the five affected instruments, not the full article/paragraph/amendment-point comparison a complete sensitivity analysis would run. Answering RQ3: neither context-window length nor parameter count predicts which model posts the best point estimate on any metric here, so embedding-model choice on a corpus like this has to be benchmarked directly rather than read off the specification sheet.

## VI. DISCUSSION

**Why does fusion not pay off here?** Fusion improves on its stronger component only if the weaker one rescues queries the stronger one misses, and that almost never happens here: dense misses only four queries and BM25 recovers just one. With so little unique lexical signal to add, fusion can at best re-rank what dense already found, while the dense-favouring weight needed to stop BM25’s errors from corrupting the ranking also suppresses that one genuine save. This is not a general argument against hybrid retrieval: reported gains typically come where the dense component is weaker or the corpus is large enough that retrievers diverge more often; on a small, terminologically dense statutory corpus a strong bi-encoder leaves BM25 too little room to help.

**Small samples mislead.** The hybrid-vs-dense direction reversed as the test set grew (dense led on MRR at  $n=22$ ; hybrid leads at  $n=53$ ) while staying non-significant at both sizes, so “fusion hurts” and “fusion helps” were both noise, including in our own earlier drafts. Legal IR benchmarks in this size range are common, and comparisons drawn from them without significance testing deserve the same suspicion.

**Evaluation design decides the apparent winner.** The sharpest result came before any system comparison: rewording the queries alone, with everything else fixed, moved BM25



far more than dense (Table III), and the independent raw-parliamentary-question check (Section IV-A) reproduces the pattern on untouched data. Annotators who write queries while reading the corpus reuse its vocabulary, so we suspect this inflation recurs in other specialised-domain benchmarks and systematically flatters lexical and hybrid systems; externally sourced information needs are a cheap, auditable defence.

**Practical implications.** These results support a simple recipe: start with a strong open-weight bi-encoder, treat BM25 fusion as low-cost insurance (negligible overhead, and on this evidence negligible benefit), and tune it if used. The literature default ( $k=60$ , equal weights) [14] degrades validation HR@5/MRR@5 from 1.000/0.810 to 0.898/0.679 because with only  $\sim 1,200$  candidates BM25’s errors displace dense rank-1 results, consistent with Bruch et al. [20]; a post-hoc full-grid sweep (Fig. 4), run to characterise sensitivity rather than re-select parameters, reproduces the deployed numbers exactly and shows HR@5 plateauing at the dense-only ceiling (0.925) from  $w_d=8$ , a broad ridge rather than a narrow spike.

Finally, the failure analysis locates the remaining headroom in encoder choice rather than corpus engineering: the largest residual failure is a cross-instrument discrimination error that another open-weight encoder resolves from the same indexed text, and no failure traces to chunking or coverage.

## VII. CONCLUSION

This paper’s most transferable lesson concerns benchmark construction rather than architecture. The sharpest single result is not a system comparison at all: with corpus, indices, systems, and relevance judgements held fixed, rewording the queries alone shifts MRR@5 by 0.29 for BM25 but only 0.09 for the dense retriever (Table III), enough to turn an apparent three-point lexical-semantic parity into a twenty-five-point gap, and the independent raw-parliamentary-question check (Section IV-A) reproduces the pattern on data that experiment never touched. Annotators who write queries while reading the corpus inherit its vocabulary, so we expect this inflation in other specialised-domain benchmarks, where it systematically flatters lexical and hybrid systems; plain-language rewriting and externally sourced information needs are a cheap, auditable defence that we argue should be standard practice when benchmarking statute-level retrieval. We set out to test whether hybrid sparse-dense fusion improves statute-level retrieval for EU Climate Law, and found that answering the question fairly required building the evaluation as carefully as the system: a reproducible 1,156-provision corpus of 72 EU climate instruments, a 102-query benchmark under an explicit leakage-control protocol with 31 test queries from documented European Parliament written questions, and a comparison of BM25, SPLADE, a pure dense bi-encoder, and weighted reciprocal rank fusion with significance testing throughout.

The findings are easy to state, with the caveats attached to each. Both dense-only and hybrid beat BM25 by wide, significant margins, and SPLADE closes most (not all) of that gap while carrying the highest hard-negative rate tested. Between dense and the hybrid no test detects a difference;

fusion is a cheap safety net, not a necessary ingredient, and we characterise the deployed system as dense-dominated weighted RRF ( $w_d=5w_s$ ), not a balanced hybrid. The embedding ablation shows no significant difference between 512-token and long-context encoders, and a targeted re-index shows no measurable effect from chunking on the five instruments it touches, so encoder identity, not context length or chunking, carries the remaining variance. We also closed a symmetry gap in our own tuning: grid-searched BM25 hyperparameters improve on validation but don’t generalise to the test set, so we kept the defaults rather than switch on an overfit signal, while stemming alone would have bought +0.057 HR@5 left on the table for exact-match precision. The remaining failures are encoder-specific discrimination and register-gap errors rather than coverage or chunking problems, so embedding-model selection, not corpus preprocessing, is a more promising next step for practitioners than it is a proven one: this rests on one demonstrated case (q080) and an aggregate parity result, not a measured comparison of remediation strategies.

Several limitations bound these conclusions. The test set, at 53 queries, yields wide confidence intervals, and a null result at this size cannot exclude a small true effect. Relevance is judged at the instrument (CELEX) level rather than the article level, which is lenient toward systems that find the right act but the wrong provision. Query paraphrases and relevance mappings were produced with generative-AI assistance and verified by the authors. A completed check by a second annotator (Faisal Ali Agha, LL.M., Trinity College Dublin, energy arbitration law) on the 14-query overlap subset found moderate agreement (Cohen’s  $\kappa=0.42$ , 95 % CI [0.20, 0.62]): below the 0.60 publication-quality bar but well above chance, indicating relevance judgements are broadly, if imperfectly, consistent across annotators without undermining the relative comparisons between systems, which use one qrel set throughout. 31 of the 53 test queries (58.5 %) still carry independent corroboration via the Commission’s official answers to the source parliamentary questions. Finally, the study covers one legal domain and one language, and all three dense models (plus SPLADE) are general-domain: no legal-domain or multilingual EU-law-specific encoder was evaluated, so “dense is enough” is a claim about strong general-domain bi-encoders on small statutory corpora, not about dense retrieval in general. `nomic-embed-text-v1.5`’s selection is also not pre-registered: version control shows it replaced two earlier candidates, each swapped for a stated technical reason before the sealed test set reached its current size, evidence against cherry-picking but not a pre-registered protocol.

Future work follows from these limitations: extend relevance judgements from the instrument to the article level; broaden the completed second-annotator check (Cohen’s  $\kappa=0.42$  on 14 queries) to the full test set and reconcile disagreements toward the 0.60 bar; address the cross-instrument discrimination failure at the encoder level via fine-tuning or re-ranking, and compare weighted RRF against a cross-encoder reranker over the same first-stage candidates; scale the test set via the European Parliament’s open data interface, which our

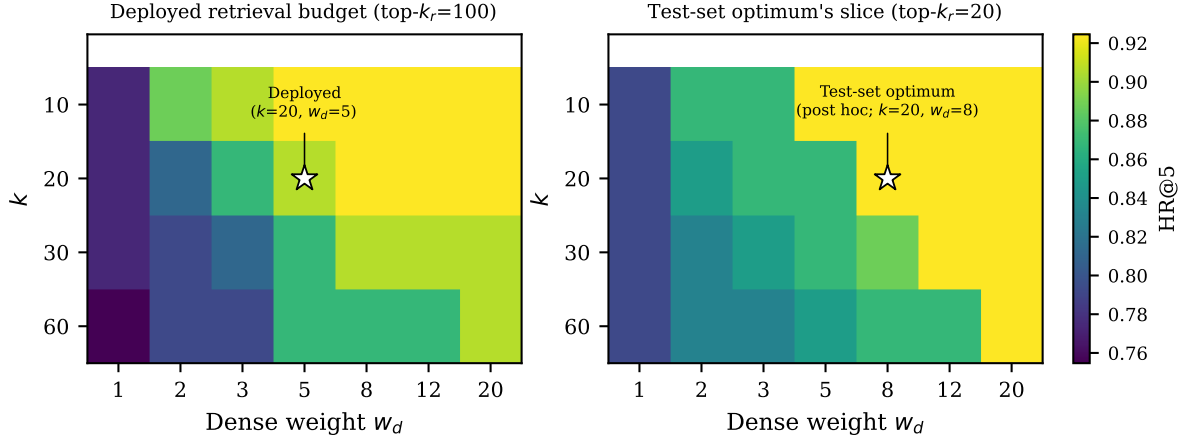


Fig. 4: HR@5 on the sealed test set across the  $(k, w_d)$  grid, from the post-hoc sensitivity sweep. Left: the deployed retrieval budget (top- $k_r=100$ ) with the deployed cell starred. Right: the slice containing the post-hoc test-set optimum (top- $k_r=20$ ). In both panels the high-scoring region is a broad plateau covering most of  $w_d \geq 5$ , not an isolated peak: the tuned configuration is not a narrow, fragile optimum.

query-derivation protocol can largely automate; and evaluate end-to-end RAG generation by feeding the top-5 retrieved passages to an LLM and scoring its answers against the Commission’s official written replies to the 31 EP-derived queries.

#### GENERATIVE AI DISCLOSURE

Generative AI tools (Claude, Anthropic) were used for proofreading and improving clarity of writing throughout; to assist in drafting the plain-language query paraphrases and candidate relevance mappings described in Section IV (Benchmark); to draft and revise expository prose in the Abstract and in Sections I (Introduction), III-C (Learned-Sparse Retrieval), V-B (Effect of Query Difficulty), and VII (Conclusion); and to compute the SPLADE and BGE-small-v1.5 rows of Table II from previously saved per-query results. All such output was reviewed and verified by the authors, and all research ideas, experimental design, analysis, and conclusions are the authors’ own.

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